STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – II

Shear Force and Bending Moment

Handwritten Study Notes

Syllabus topics covered in this unit :

Definition of beam — Types of beams — Concept of shear force and bending moment — Point of contra flexure — Relation between S.F., B.M. and rate of loading — S.F. and B.M. diagrams for cantilever, simply supported and overhanging beams subjected to point loads, U.D.L., U.V.L., partial U.D.L., couple and combinations

Contents of these notes :

→ 2.1 Beams – Definition and Types
→ 2.2 Shear Force and Bending Moment
→ 2.3 Cantilever Beams
→ 2.4 Simply Supported Beams
→ 2.5 Overhanging Beams

2.1 Beams – Definition and Types

A beam is a structural member, usually horizontal, that carries loads transverse (perpendicular) to its axis and resists them by bending and shear.

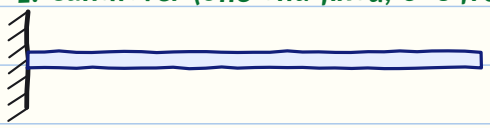
- Loads cause internal shear force (S.F.) and bending moment (B.M.) – these decide the section size.

Types of supports

- **Simple / roller support** : gives one vertical reaction ; beam free to rotate and slide.
- **Hinged (pin) support** : gives vertical + horizontal reaction ; rotation free.
- **Fixed support** : gives V, H reactions and a fixing moment ; no rotation or translation.

Types of beams

1. Cantilever (one end fixed, one free)



2. Simply supported



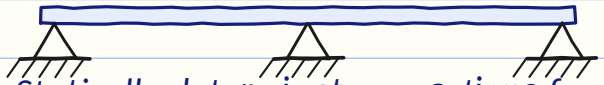
3. Overhanging (one/both ends)



4. Fixed (built-in) beam



5. Continuous beam (3 or more supports)



Propped cantilever = fixed + prop at free end

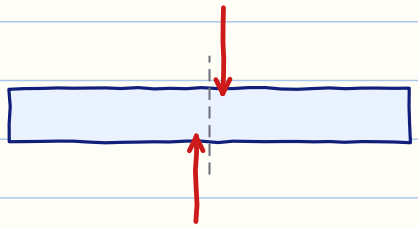
Determinate: 1, 2, 3 (statics only)

- Statically determinate : reactions found by $\Sigma V = 0$, $\Sigma H = 0$, $\Sigma M = 0$ (cantilever, simply supported, overhanging).
- Statically indeterminate : extra reactions – fixed, propped cantilever, continuous beams.

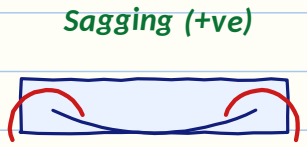
Types of loads

- Point (concentrated) load P – kN, acts at a point.
- Uniformly distributed load (U.D.L.) w – kN/m ; equivalent point load $W = w\,L$ at the centre.
- Uniformly varying load (U.V.L.) – intensity varies linearly ; triangular load $W = \frac{1}{2}\,w\,L$ acting at $L/3$ from the larger end.
- Couple / moment M – kN·m, produces rotation only.

+ve Shear : left part ↑, right part ↓

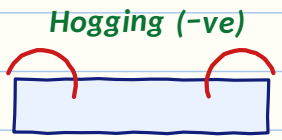


(clockwise shear on the segment)



Sagging (+ve)

concave upward



Hogging (−ve)

concave downward

2.2 Shear Force and Bending Moment

Shear force at a section = algebraic sum of all vertical forces to ONE side of the section.

Bending moment at a section = algebraic sum of moments of all forces to ONE side of the section about that section.

Sign conventions (sagging +ve)

- S.F. : +ve if the left part tends to move UP (and right part DOWN) – clockwise shear.
- B.M. : +ve (sagging) when it bends the beam concave upward ; −ve (hogging) when concave downward.
- Quick rule : forces/moments to the LEFT : upward force +ve, clockwise moment +ve ; to the RIGHT : reverse.

Relation between load, S.F. and B.M.

Consider an element dx with load intensity w (down) :

$$\frac{dF}{dx} = -w \qquad \frac{dM}{dx} = F$$

$$\frac{d^2M}{dx^2} = -w$$

- Slope of S.F. diagram = −(load intensity).
- Slope of B.M. diagram = S.F. at that section.
- Change in S.F. between two sections = − area of load diagram.
- Change in B.M. between two sections = area of S.F. diagram.
- Maximum B.M. occurs where S.F. = 0 (or changes sign).

Point of contra-flexure (P.C.)

The point where the bending moment changes sign (B.M. = 0 and curvature reverses). Occurs only in overhanging / fixed / continuous beams.

Shapes of diagrams

Loading on span	S.F.D.	B.M.D.
No load	horizontal line	inclined line
Point load	sudden jump = load	kink (slope changes)
U.D.L.	inclined line	parabola
U.V.L.	parabola	cubic parabola
Couple	no change	sudden jump = couple

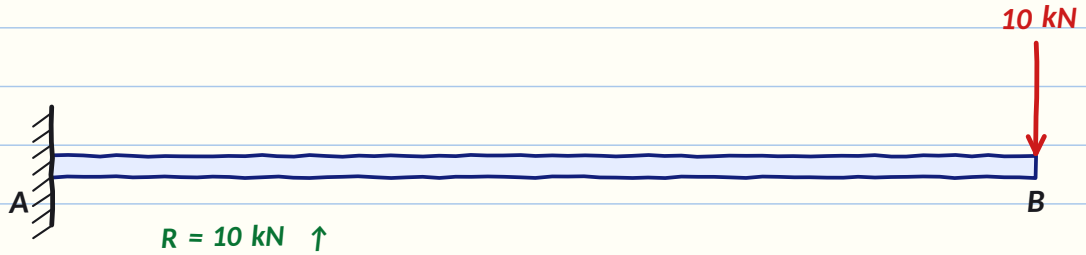
Procedure for drawing S.F.D and B.M.D

- 1. Replace distributed loads by equivalent point loads ; find reactions.
- 2. Divide the beam into segments at loads/supports.
- 3. Find F and M just left/right of every key section ; plot to scale.
- 4. Join using the shape rules ; locate F = 0 for M_{max} ; find P.C.

2.3 Cantilever Beams

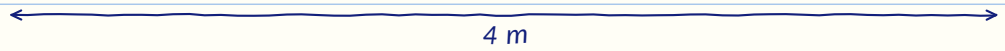
Free-end at B, fixed at A. Take x from the free end – no need to find reactions.

Case 1 : Point load W at free end (L = 4 m, W = 10 kN)

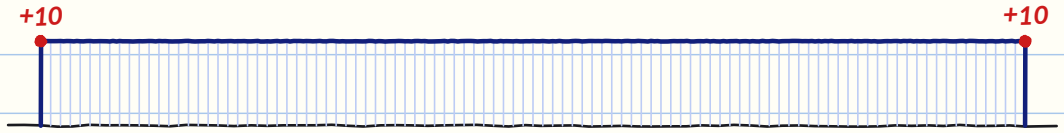


$R = 10$ kN ↑

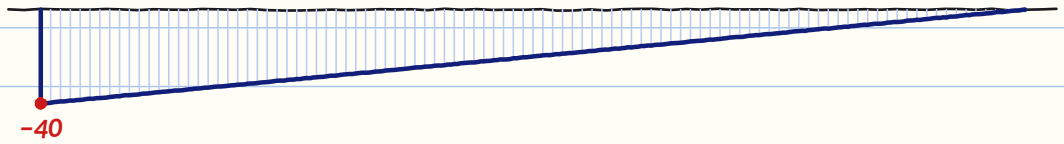
$M_A = 40$ kN·m (fixed-end moment)



S.F.D

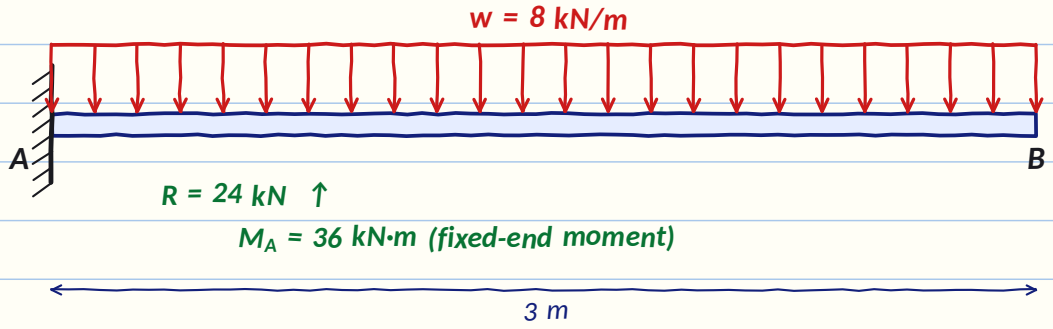


B.M.D

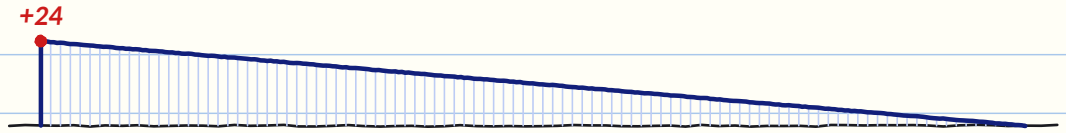


- $F_x = +W$ (constant) $= +10$ kN ; $M_x = -Wx$; $M_A = -WL = -10 \times 4 =$ **–40 kN·m**

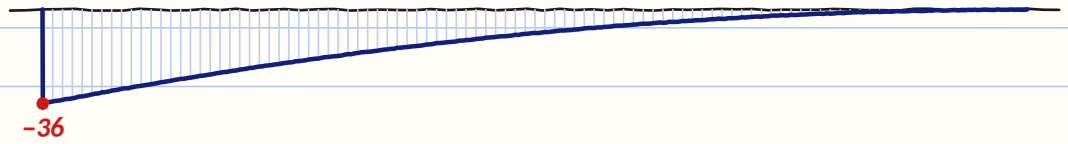
Case 2 : U.D.L. w over full span (L = 3 m, w = 8 kN/m)



S.F.D



B.M.D



- $F_x = w\,x$: $F_B = 0$, $F_A = w\,L$ = **24 kN** (straight line)
- $M_x = -\,w\,x^2/2$: $M_A = -\,8 \times 9/2$ = **–36 kN·m** (parabola)

Case 3 : U.V.L. – zero at free end, w at fixed end

- Total load $W = ½\,w\,L$ (acts at L/3 from the fixed end).

$$F_A = \frac{w\,L}{2} \qquad M_A = -\frac{w\,L^2}{6}$$

- $F_x = w\,x^2/2L$ (parabola) ; $M_x = -\,w\,x^3/6L$ (cubic).

Example 1 (Combination)

Cantilever 3 m : 10 kN at free end B, U.D.L. 4 kN/m over the whole span, couple 6 kN·m (clockwise) at mid-span.

F_B = 10 ; at mid : 10 + 4×1.5 = 16 kN ; at A : 10 + 4×3 = **22 kN**

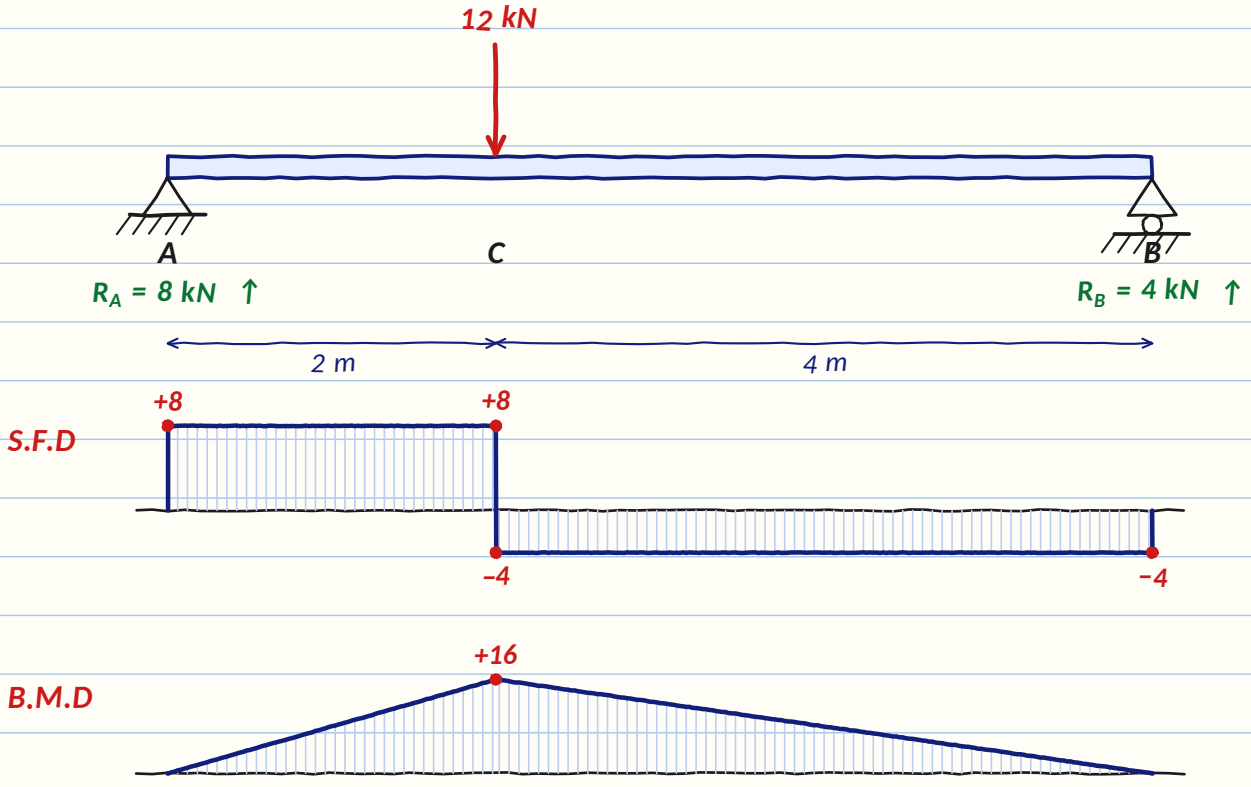
M just right of mid (from free end) = – (10×1.5 + 4×1.5×0.75) = – 19.5 kN·m

just left of mid = – 19.5 – 6 = **– 25.5 kN·m** (a clockwise couple on the right part is –ve)

M_A = – (10×3 + 4×3×1.5 + 6) = **– 54 kN·m** (hogging)

2.4 Simply Supported Beams

Case 4 : Point load (L = 6 m, W = 12 kN at C, 2 m from A)



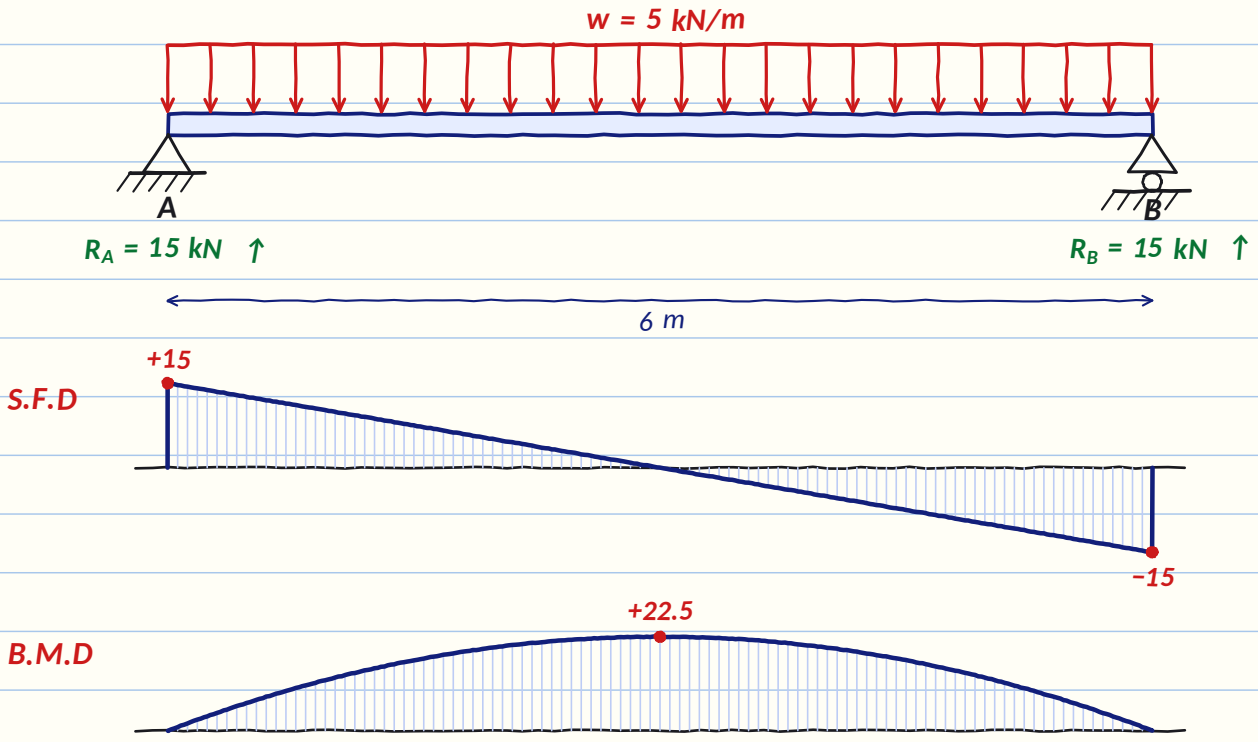
$R_B = 12 \times 2 / 6 =$ **4 kN** ; $R_A = 12 - 4 =$ **8 kN**

F : A to C = +8 ; C to B = 8 – 12 = –4 kN

$M_C = 8 \times 2 =$ **16 kN·m** (max, sagging) ; $M_A = M_B = 0$

- Central point load : $R = W/2$; $M_{max} = W\,L/4$.

Case 5 : U.D.L. over full span (w = 5 kN/m, L = 6 m)

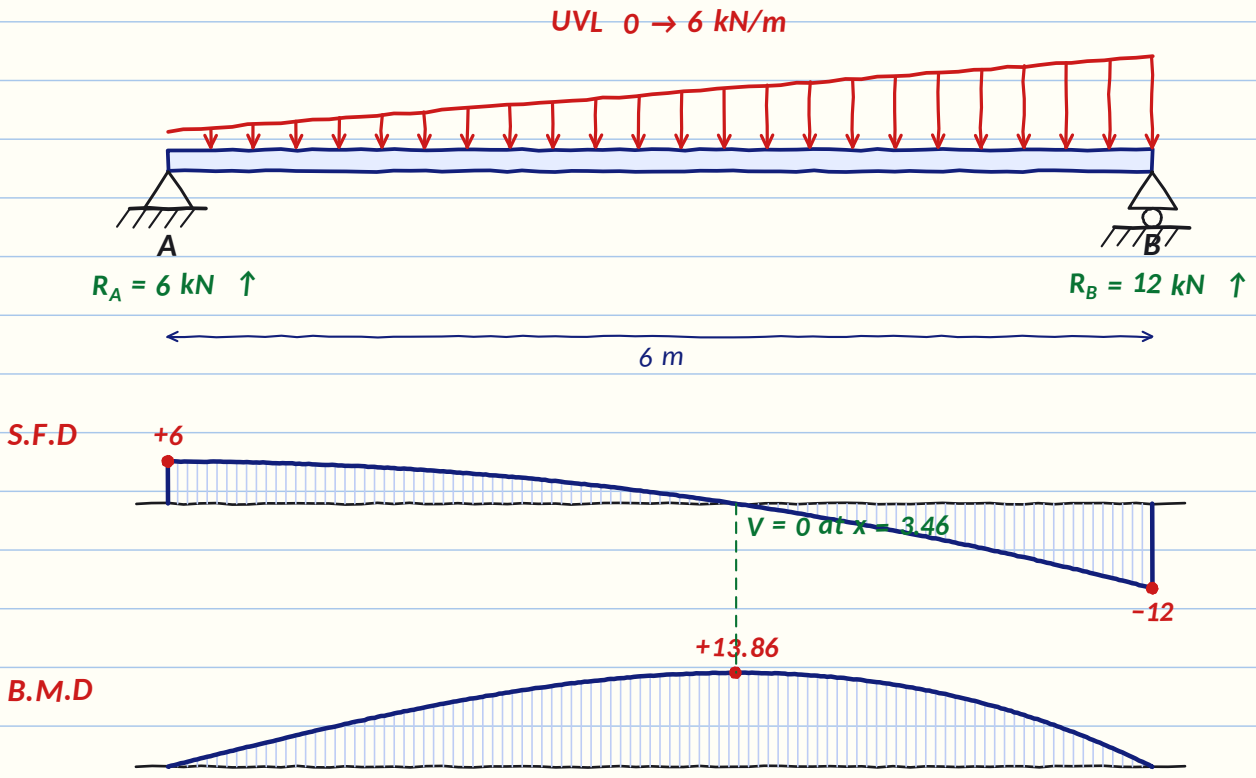


$R_A = R_B = w\,L/2 = 5 \times 6/2 =$ **15 kN**

$F_x = 15 - 5x$; $F = 0$ at $x = 3$ m (centre)

$M_{max} = w\,L^2/8 = 5 \times 36/8 =$ **22.5 kN·m** at centre (parabola)

Case 6 : U.V.L. (0 at A to 6 kN/m at B , L = 6 m)



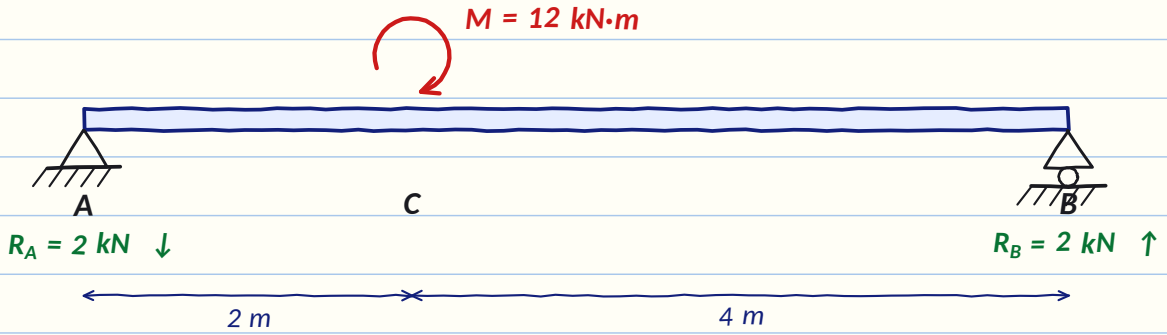
Total W = ½ × 6 × 6 = 18 kN ; $R_A = W/3$ = **6 kN** ; $R_B = 2W/3$ = **12 kN**

$F_x = 6 - x^2/2$; F = 0 at $x = L/\sqrt{3}$ = **3.464 m** from A

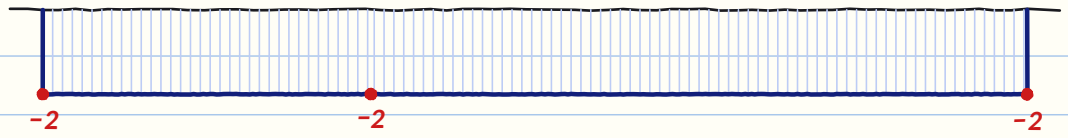
$M_x = 6x - x^3/6$; $M_{max} = 6(3.464) - 3.464^3/6$ = **13.86 kN·m**

- General : $M_{max} = 0.0642\ w\ L^2$ at 0.577 L from the thin end.

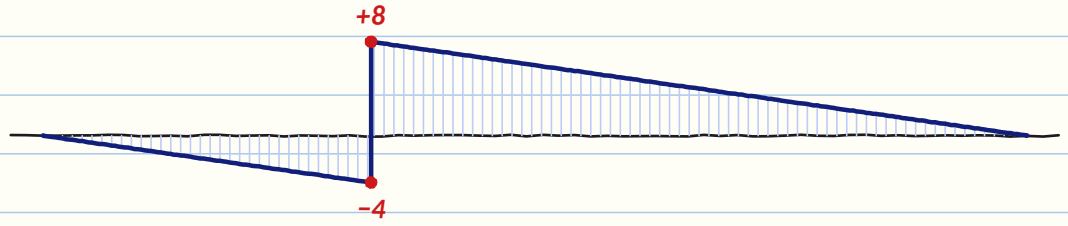
Case 7 : Couple on a simply supported beam (12 kN·m clockwise at C, 2 m fr



S.F.D



B.M.D



Take moments about A : $R_B \times 6 = 12 \Rightarrow R_B = 2$ kN (↑) ; $R_A = -2$ kN (i.e. 2 kN ↓)

S.F. = − 2 kN throughout (constant)

M just left of C = − 2 × 2 = **−4 kN·m** ; just right = −4 + 12 = **+8 kN·m**

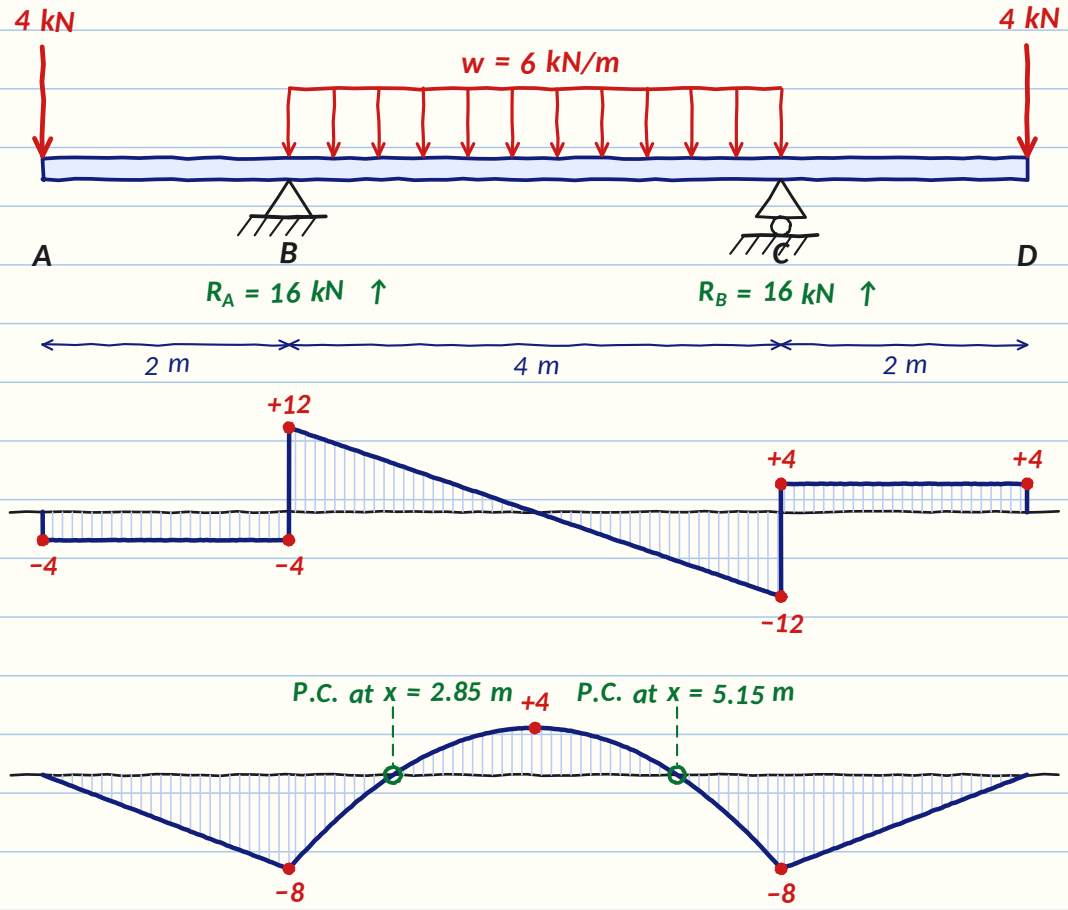
- A couple does not affect S.F. but makes the B.M. jump by its magnitude.

2.5 Overhanging Beams

Beam rests on two supports and extends beyond one or both. The B.M. changes sign ⇒ **points of contra-flexure**.

Case 8 : Overhang on both sides

Supports at B (2 m from A) and C (6 m from A) ; 4 kN at each free end A, D ; U.D.L. 6 kN/m between B and C.



By symmetry : $R_B = R_C = (4 + 4 + 24)/2 =$ **16 kN**

S.F. : A–B = −4 ; B right = −4 + 16 = +12 ; at C left = 12 – 24 = −12 ; C right = −12 + 16 = +4 ; C–D = +4

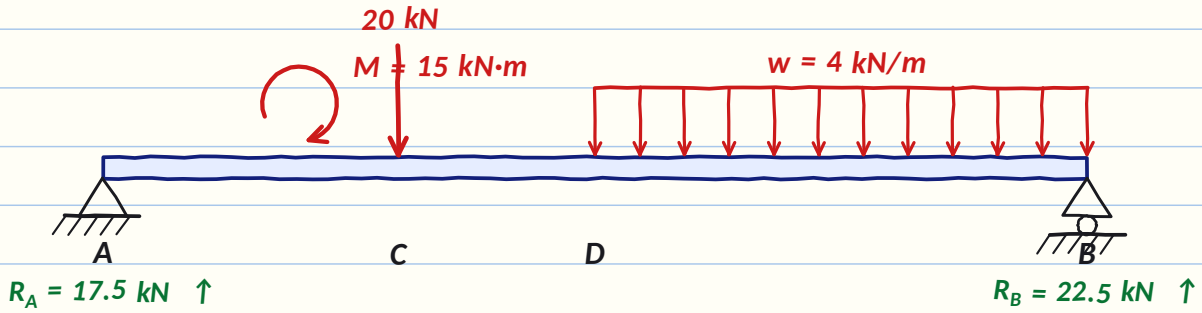
B.M. : $M_B = -4 \times 2 =$ **−8 kN·m** ; $M_C = -4 \times 2 = -8$ kN·m

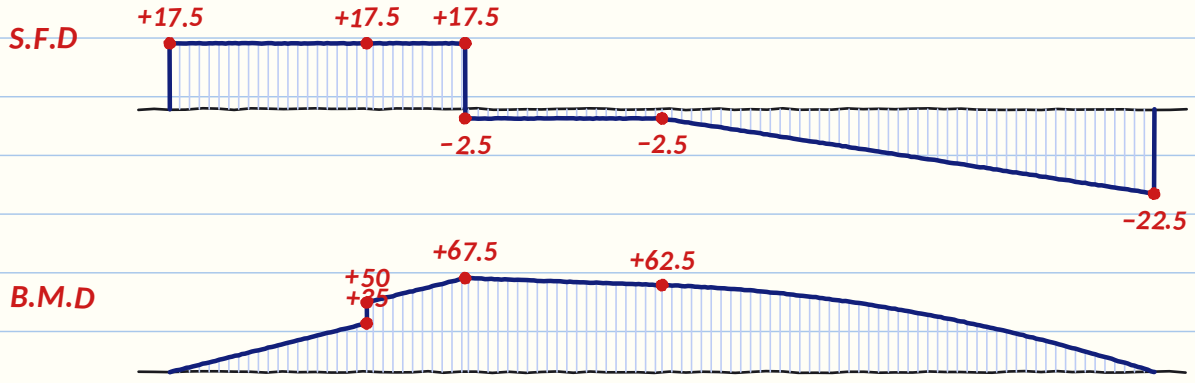
F = 0 at 2 m right of B ⇒ $M = -8 + 12\times2 - 6\times2^2/2 =$ **+4 kN·m** (max sagging, midspan)

Contra-flexure : $-8 + 12s - 3s^2 = 0 \Rightarrow s = 0.845$ m or 3.155 m from B ⇒ **x = 2.845 m and 5.155 m from A**

Case 9 : Combination (point load + partial U.D.L. + couple)

SS beam 10 m : 20 kN at 3 m from A, U.D.L. 4 kN/m from 5 m to B (10 m), clockwise couple 15 kN·m at 2 m from A.





Moments about A : $R_B \times 10 = 20\times3 + (4\times5)\times7.5 + 15 = 60 + 150 + 15 = 225$

R_B = **22.5 kN** ; $R_A = 40 - 22.5 =$ **17.5 kN**

S.F. : A–C(2) 17.5 ; 2–3 : 17.5 ; right of P : −2.5 ; at 5 m : −2.5 ; at B : −2.5 − 20 = −22.5

B.M. : $M_{x=2}$ left = 35 ; right = 35 + 15 = 50 ; $M_{x=3} = 17.5\times3 + 15 =$ **67.5 kN·m** (max, S.F. changes sign)

$M_{x=5} = 17.5\times5 + 15 - 20\times2 = 62.5$; $M_B = 0$ (check ✓)

Standard results – quick table

Beam / load	Max S.F.	Max B.M.
Cantilever, W at end	W	W L
Cantilever, w/unit length	w L	w $L^2/2$
Cantilever, U.V.L. (0 at free end)	w L/2	w $L^2/6$
S.S., W at centre	W/2	W L/4
S.S., W at distance a from A	W b/L	W a b/L
S.S., U.D.L.	w L/2	w $L^2/8$
S.S., U.V.L. (0→w)	w L/3	0.0642 w L^2

Important points to remember

- Draw S.F.D and B.M.D below the loaded beam, same scale for length.
- Always check: sum of reactions = total load ; M at free/pinned end = 0.
- Max B.M. lies at F = 0 – equate $F_x = 0$ to get its location.
- For an overhanging beam, M changes sign – find P.C. by $M_x = 0$.
- Mark +/– signs, units and key values on every diagram.